ARCHIT AGRAWAL

San Francisco Bay Area | 623-312-0435 | architagrawal000@gmail.com | LinkedIn | GitHub | Personal Website

PROFESSIONAL EXPERIENCE

EdPlus **05/2026 - Present**
AI Software Engineer *Phoenix, AZ*

- Own the architecture, roadmap and delivery of a 12-agent AI analytics platform on AWS Bedrock and Step Functions that turns any structured dataset into cited, verified answers; TypeScript, NestJS, Nuxt.
- Engineered the storage layer: kilobyte fact tables beside Lance payloads on S3 queried in-process by DuckDB, choosing Lance over Parquet on 114,000 rows for full-text and vector search.
- Scaled labeling as a distributed Step Functions fan-out across 64 parallel lanes with a 5% failure circuit breaker, after tracing silent row loss in a run that reported success.
- Detected table shape with 98% accuracy and survey weights in 100% of files that ship one, up from 0%, so crosstabs, long tables and weighted surveys publish with no per-file code.
- Accelerated publishing of a 51,280-person, 220-column federal survey 10x, 19 minutes to 109 seconds, with an independent audit from raw data matching 100% of figures.
- Eliminated a silent failure mislabeling 1 in 8 rows in the prior pipeline, reaching 100% join integrity with schema-constrained LLM output.
- Unlocked a time axis on roughly 70% of real exports that hid timestamps inside identifiers, through a five-stage intake that also survives UTF-16 and duplicate-header files.
- Blocked a cross-survey join that double counted respondents, using a fingerprint that detects runs sharing the same people.
- Sped up a full AWS pipeline run 6x, 500 to 85 seconds, by replacing managed agent-runtime stages with direct tool calls once benchmarks showed each made one call and stopped.
- Lowered labeling spend by testing codebook fit on a 25-row sample first, separating a workable codebook (0.598 fit) from an unusable one (0.316) for about $0.001.
- Cut wrong answers 85% on 164 published benchmark figures by reading every answer back in plain words and verifying it against the question before display.
- Extended analysis to statistical and causal questions with 25 tests and six causal estimators validated against scipy and statsmodels, scoring 68% on StatQA versus GPT-4o's best reported 64.83%.
- Delivered governance in the POC: Bedrock Guardrails as their own CDK stack, entitlement-gated respondent retrieval, and per-stage spend caps enforced in code.
- Authored the target production architecture: 80 logged decisions and 200+ measured experiments, grounded in prior art including SSSOM, DDI, Metabase MBQL, Vega-Lite, Draco and LIDA.
- Reset the team's accuracy baseline after tracing a reported 0.637 F1 to a hand-reviewed spreadsheet; fresh runs reproduce 0.523.
- Quadrupled the insights automated reports surface, 13% to 52–59% on unseen datasets, by having an LLM agent choose the analyses while deterministic code computes every number.

MyStage Music Inc. **07/2025 - 06/2026**
Founding AI/ML Engineer *Remote*

- Scaled ingestion to 70,000+ records a day from 650+ venues with a Playwright pipeline, transactional locking and Gemini entity resolution that raised search-dataset accuracy 25%.
- Guaranteed retry safety with deterministic task IDs and create-if-absent writes, so a retried job can never overwrite one already in flight.
- Re-architected a 14-Cloud-Function backend into one checkpointed LangGraph agent in Python on GCP Cloud Run, cutting production service dependencies 64%.
- Delivered sub-50 ms search on FastAPI and Algolia with real-time Firestore sync, 200+ pytest tests and Logfire tracing.
- Designed a reusable scraping subgraph that parent graphs compose without checkpoint conflicts, letting webhook-triggered scrapes skip the fetch step.
- Implemented pause-and-resume on LangGraph interrupts, so a record missing context waits at its checkpoint instead of failing the batch.

EdPlus **09/2023 - 05/2025**
Software Engineer *Phoenix, AZ*

- Cut transcript analysis 16x, four hours to 15 minutes, with a Neo4j knowledge graph queried by validated LLM-written Cypher.
- Isolated each college's data at the retrieval layer, so no tenant's material can surface in another's and onboarding became a config change, not a deployment.
- Led a multi-tenant RAG assistant, React to retrieval layer, used by 1,000+ faculty to build courses for 60,000+ students.
- Compared fixed-size against section-aware chunking on real course documents before choosing, because a split policy paragraph answers half a question.

- Added hybrid keyword and vector search so exact course codes resolve correctly where embeddings alone blur them together.

Knowledge Exchange for Resilience, Arizona State University **06/2024 - 08/2024**
Data Research Aide *Phoenix, AZ*

- Built a FastAPI and pgvector recommender reaching 85% top-k precision for collaborator matching, fed by an ETL ingesting 10,000+ profiles a day, plus 20+ REST APIs on .NET 8.
- Cut p95 retrieval latency 60% at 95% recall with a hybrid FAISS and Neo4j search layer.
- Rewrote the ETL to be idempotent on a natural key, so a rerun after partial failure updates rows rather than duplicating them.
- Raised dataset accuracy to 95% by enforcing uniqueness, not-null and range checks in PostgreSQL, so bad rows fail at the boundary instead of in a report.

Zeus Learning **01/2022 - 07/2023**
Software Engineer *Mumbai, India*

- Migrated a C# .NET monolith to Dockerized Kubernetes microservices, reducing infrastructure cost 20% and footprint 35%.
- Cut deploy time 70% and production incidents 40% by containerizing the CI/CD release pipeline and adding SonarQube gates.
- Sped up the student-listing screen 30% with paginated fetching and S3 asset delivery, then another 10% with index-covering query rewrites after profiling found a full table scan.
- Right-sized Kubernetes requests and limits to observed usage, which is where the 35% footprint reduction came from.

EAT.FIT **09/2021 - 12/2021**
Product Intern *Bengaluru, India*

- Cut driver-tracking server cost 45% with distance-based polling and client-side interpolation that kept map movement smooth.
- Built a WebSocket and Express notification service with backpressure-aware fan-out and duplicate-safe redelivery; reported CSAT rose 28% and support volume fell 35%.

DA-IICT **05/2021 - 08/2021**
Computer Vision Researcher *Gandhinagar, India*

- Engineered an autonomous-driving perception stack in Python: radar and lidar capture, timestamp alignment, extrinsic calibration and point-cloud processing, with a replay harness over recorded drives.
- Compared early and late sensor fusion on identical sequences, showing point-level merging keeps detail but inherits both sensors' noise while object-level merging is robust but discards evidence.

PROJECTS

SRP Electric MCP Server | GitHub **12/2025 - 01/2026**

- Built a Python MCP server on FastMCP that logs into an undocumented utility portal and gives AI agents 7 tools and 4 analysis prompts for billing, usage, payments and outages, with a one-click setup wizard.
- Added analysis prompts for usage trends, bill explanations, savings advice and account comparison on top of the raw data tools.

MCP GitHub PR Review Agent **07/2025 - 08/2025**

- Built a TypeScript MCP service that reviews pull requests from the diff, linked ticket and CI result, returning schema-typed inline findings that gate merges by category.

AiJockey – AI DJ Pipeline | GitHub **2025 - Present**

- Built an end-to-end AI DJ in PyTorch and FastAPI: stem separation, an LLM transition planner, 25+ DSP transitions and LUFS mastering, ported off CUDA to an AMD MI300X.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#
Frameworks: FastAPI, NestJS, Node.js, .NET, React, Vue/Nuxt, Next.js, LangGraph, PyTorch
AI & Data: RAG, LLM evaluation, MCP, AWS Bedrock, PostgreSQL, pgvector, FAISS, Neo4j, Redis, DuckDB, ETL pipelines
Cloud & DevOps: AWS (Step Functions, Lambda, S3, CDK), GCP Cloud Run, Docker, Kubernetes, CI/CD

EDUCATION

Arizona State University (ASU) **08/2023 - 05/2025**
Master of Science, Computer Science *Tempe, AZ, USA*

- **GPA:** 4.0

Dhirubhai Ambani Institute of Information and Communication Technology **08/2018 - 05/2022**
Bachelor of Technology, Information and Communication Technology *India*